SYSTEM REQUIREMENTS SPECIFICATION

Smart Home Security System

Design and Validation Project

Document type	System Requirements Specification (SysRS)
Structure	ISO/IEC/IEEE 29148:2018 - SysRS section 9.4
System	Smart Home Security System
Architecture	Local-first hybrid edge + cloud
Visual theme	Gray and orange
Version / date	Final project version - October 2026

Design focus

I designed the system so that essential home-security behavior remains local. Cloud services improve remote access and resilience, but they are not required for basic intrusion detection, alarm activation, or local event logging.

Contents

1. Introduction
    1.1 Purpose
    1.2 Scope
2. Overview
    2.1 System Context
    2.2 System Functions
    2.3 User Characteristics
3. Functional Requirements
4. Usability Requirements
5. Performance Requirements
6. System Interfaces
7. System Operations
    7.1 Human System Integration Requirements
    7.2 Maintainability
    7.3 Reliability
8. System Modes and States
    8.1 Physical Requirements
    8.2 Adaptability Requirements
    8.3 Environmental Conditions
    8.4 System Security
9. Verification
10. Assumptions and Dependencies
Appendix A - Requirement Traceability Summary
References

1. Introduction

1.1 Purpose

The Smart Home Security System (SHSS) is intended to help homeowners and residents monitor a residential property, detect security or safety events, and respond quickly through local alarms and user notifications. The system combines sensors, cameras, smart-device controllers, a local control unit, user interfaces, local storage, and optional cloud services.

This SysRS defines the expected system behavior and the measurable qualities that will guide architecture, component selection, integration, verification, validation, maintenance, and future expansion. Requirements use unique identifiers so that design decisions and tests can be traced back to the specification.

1.2 Scope

System name: Smart Home Security System (SHSS).

The SHSS covers residential monitoring and control for intrusion detection, door/window status, water-leak detection, video surveillance, local alarms, secure remote access, event history, and selected smart-home actions such as lock and lighting control.

The system is expected to:

- **SCOPE-01 -** Detect motion and protected-entry events from registered sensors.
- **SCOPE-02 -** Detect water leaks and present safety alerts.
- **SCOPE-03 -** Provide live and event-related video from registered cameras.
- **SCOPE-04 -** Allow authorized users to arm, disarm, monitor, and review security information.
- **SCOPE-05 -** Keep essential local security behavior available during Internet loss.
- **SCOPE-06 -** Use cloud services for remote access, notifications, synchronization, and backup where configured.

Boundary of the project
The SHSS does not itself provide police, fire, medical, or professional monitoring services, and it cannot guarantee prevention of intrusion or property damage. Those services would require separate external integrations and agreements.

2. Overview

2.1 System Context

The central control unit is the local decision point. It receives events, applies configured security rules, controls alarms and actuators, stores event information, and exchanges data with the user interfaces and cloud services. Low-power sensors and controllers use a Zigbee mesh; cameras use the local IP network; remote user access uses encrypted network interfaces.

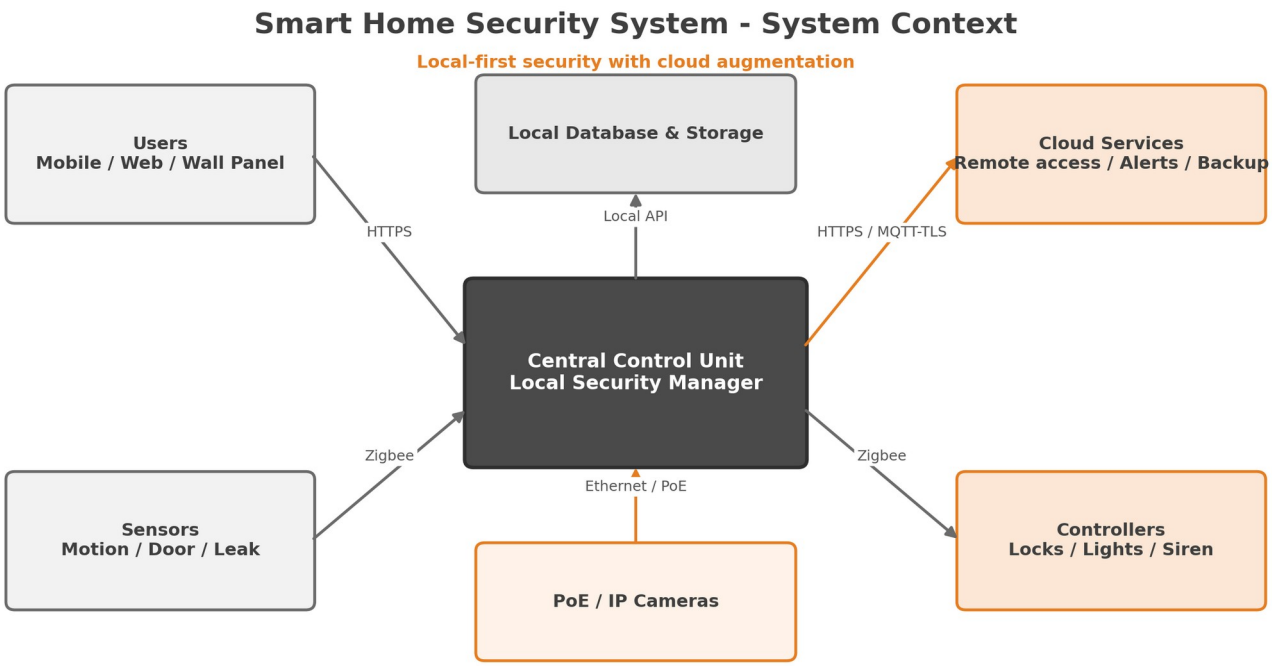


Figure 1. System context and major interfaces.

My architecture choice

The local-first hybrid approach is deliberate: sensor processing and alarm behavior stay close to the home, while cloud services add convenience. This reduces dependence on Internet availability for time-critical security behavior.

2.2 System Functions

ID	System function
SF-01	Detect motion in monitored areas.
SF-02	Detect open/close changes at protected doors and windows.
SF-03	Detect water leaks.
SF-04	Capture and stream security-camera video.
SF-05	Activate local audible or visual alarms.
SF-06	Send alerts to authorized users.
SF-07	Arm and disarm the security system.
SF-08	Authenticate users and enforce role permissions.
SF-09	Provide secure remote monitoring and control.
SF-10	Store security events and related metadata.
SF-11	Control compatible locks, lights, and alarm actuators.
SF-12	Continue essential local functions during Internet loss.

2.3 User Characteristics

User type	Typical number	Characteristics and permitted use
Homeowner / Administrator	1-2	Full configuration authority. Manages users, devices, rules, event history, camera permissions, arm/disarm behavior, and system settings.
Resident	1-8	Routine household user. Receives alerts, views permitted cameras, checks system state, and arms/disarms allowed zones.
Installer / Maintenance Technician	0-1 active at a time	Technical user who installs, pairs, diagnoses, updates, and replaces components. Access to private household information is limited to the maintenance task.

3. Functional Requirements

The word "shall" identifies a mandatory system requirement. Each requirement has a stable ID for traceability to design and test evidence.

Motion Detection

- **FR-MOT-01 -** The system shall accept motion events from registered motion sensors.
- **FR-MOT-02 -** When motion occurs in an armed zone, the system shall create a security event.
- **FR-MOT-03 -** The system shall notify authorized users when an armed motion sensor triggers.
- **FR-MOT-04 -** Each motion event record shall include the triggering device, zone, and timestamp.

Door and Window Monitoring

- **FR-DW-01 -** The system shall detect open and close reports from registered door/window sensors.
- **FR-DW-02 -** Opening a protected entry point while its zone is armed shall create an intrusion event.
- **FR-DW-03 -** The event record shall identify the protected entry point and timestamp.

Water Leak Detection

- **FR-WL-01 -** The system shall detect leak events from registered water sensors.
- **FR-WL-02 -** Authorized users shall be notified when a leak is detected.
- **FR-WL-03 -** A leak alert shall remain active until acknowledged or until the sensor returns to normal, according to configuration.

Video Surveillance

- **FR-CAM-01 -** The system shall support live video from registered IP cameras.
- **FR-CAM-02 -** Authorized users shall be able to view permitted camera feeds remotely.
- **FR-CAM-03 -** Configured security events shall be able to trigger recording.
- **FR-CAM-04 -** Recorded-video metadata shall be linked to the related security event.
- **FR-CAM-05 -** The system shall indicate when a registered camera becomes unavailable.

Alarm System

- **FR-ALM-01 -** The system shall activate the configured local alarm when a qualifying intrusion rule is met.
- **FR-ALM-02 -** The alarm subsystem shall support an audible warning device.
- **FR-ALM-03 -** Authorized users shall receive an alert when the local alarm is activated.
- **FR-ALM-04 -** An authorized user shall be able to silence or reset the alarm according to role permissions.

Central Control Unit

- **FR-CCU-01 -** The control unit shall receive and process events from registered sensors and controllers.
- **FR-CCU-02 -** The control unit shall execute configured security and automation rules.

- **FR-CCU-03 -** The control unit shall continue essential detection, local alarm handling, and event logging without cloud connectivity.
- **FR-CCU-04 -** The control unit shall maintain registration and health-state information for connected devices.

User Authentication and Authorization

- **FR-AUTH-01 -** Users shall authenticate before accessing security controls or private event data.
- **FR-AUTH-02 -** Invalid credentials shall be rejected.
- **FR-AUTH-03 -** The system shall provide at least administrator and resident roles.
- **FR-AUTH-04 -** Inactive authenticated sessions shall end after a configurable timeout.
- **FR-AUTH-05 -** Administrative configuration functions shall be restricted to authorized administrator accounts.

Smart-Device Control

- **FR-CTL-01 -** The system shall support compatible smart locks, smart lights, and siren controllers.
- **FR-CTL-02 -** The local control unit shall be able to issue commands to registered compatible controllers.
- **FR-CTL-03 -** The user interface shall display the last known state of registered controllable devices.

4. Usability Requirements

Usability requirements

- **UR-01 -** After authentication, a resident shall be able to arm or disarm the system in no more than three interface actions.
- **UR-02 -** The primary interface shall clearly display the current state: Disarmed, Armed Stay, Armed Away, Alert, Alarm, Maintenance, or Offline/Degraded.
- **UR-03 -** Security alerts shall identify the event type, triggering device or location, and event time.
- **UR-04 -** At least 90% of representative first-time users shall successfully arm and disarm the system without operational assistance.
- **UR-05 -** Critical alarm actions shall use clear labels and provide confirmation feedback.
- **UR-06 -** Recoverable errors shall explain the failed action and provide a user-relevant next step.

5. Performance Requirements

Performance requirements

- **PR-01 -** A normal sensor event shall be processed and reflected in local system state within 1 second of receipt.
- **PR-02 -** A configured local alarm shall activate within 2 seconds after confirmation of a qualifying intrusion event.
- **PR-03 -** When Internet access is available, an alarm notification shall be submitted to the remote notification service within 3 seconds of local alarm activation.
- **PR-04 -** The system shall support at least 50 registered sensors and controllers.
- **PR-05 -** The system shall support at least 8 registered IP cameras, subject to available bandwidth and storage.
- **PR-06 -** At least 5 authorized user devices shall be able to remain connected at the same time without loss of critical control functionality.
- **PR-07 -** The local control unit shall support continuous 24-hour operation under normal residential conditions.
- **PR-08 -** After a normal software restart, the control unit shall restore essential service and security-state information within 120 seconds.

6. System Interfaces

The interface design separates low-power device traffic, high-bandwidth video, local control, and remote-user communication. This separation also supports the tradeoff decisions documented in the decision-matrix workbook.

Interface	Primary technology	Direction / purpose
Motion / door / leak sensors	Zigbee mesh	Sensor events to the local control unit; pairing and device management in both directions.
Smart locks / lights / siren controllers	Zigbee	Local commands from the control unit with

Interface	Primary technology	Direction / purpose
		returned state/status.
Security cameras	Ethernet/PoE or approved local IP	Live video, event clips, and camera-health information.
Mobile/Web UI	HTTPS	Authenticated requests, status, alerts, configuration, and camera access.
Control unit to cloud	HTTPS and/or MQTT over TLS	Notifications, remote state synchronization, and optional backup.
Control unit to local storage	Local API / service	Events, configuration, audit records, and video metadata.
Power subsystem	AC mains with backup where required	Power for hub/network/cameras; battery power for selected sensors.

Interface requirements

- **IF-01 -** Low-power wireless sensors shall use a supported Zigbee-compatible interface.
- **IF-02 -** IP cameras shall communicate through the local IP network using Ethernet/PoE or approved Wi-Fi connectivity.
- **IF-03 -** Remote user communication shall use encrypted HTTPS.
- **IF-04 -** Cloud messaging, when used, shall use encrypted transport such as HTTPS or MQTT over TLS.
- **IF-05 -** Loss of a cloud or remote interface shall not disable essential local intrusion detection or alarm processing.

7. System Operations

7.1 Human System Integration Requirements

Human-system integration

- **HSI-01 -** Arm/disarm, alarm acknowledgement, and emergency alarm status shall be directly visible from the primary interface.
- **HSI-02 -** High-impact administrative actions such as deleting a user or disabling an alarm rule shall require confirmation.
- **HSI-03 -** Maintenance access shall be role-restricted and shall not require access to unrelated private video or household information.
- **HSI-04 -** Normal, warning, and alarm states shall use text labels in addition to color so that meaning is not conveyed by color alone.

7.2 Maintainability

Maintainability requirements

- **MR-01 -** A failed supported sensor shall be replaceable without replacing the central control unit.
- **MR-02 -** The system shall expose health and connectivity status for registered devices.
- **MR-03 -** Supported software and firmware updates shall not require complete system replacement.
- **MR-04 -** A trained technician shall be able to replace and re-register a supported sensor within 30 minutes under normal installation conditions.
- **MR-05 -** Configuration backup shall be sufficient to restore core settings after local storage replacement or corruption.
- **MR-06 -** Diagnostic logs shall contain timestamped component and communication errors useful for troubleshooting.

7.3 Reliability

Reliability requirements

- **REL-01 -** Essential intrusion detection and local alarm processing shall continue during Internet loss.
- **REL-02 -** The system shall automatically attempt to reconnect to cloud services after connectivity returns.
- **REL-03 -** The local control unit shall provide at least 99.5% operational availability per calendar month, excluding planned maintenance.

- **REL-04 -** Critical security events created while cloud connectivity is unavailable shall be retained locally until synchronization is possible or local retention limits are reached.
- **REL-05 -** Failure of one non-critical sensor shall not stop other registered sensors from reporting events.
- **REL-06 -** The system shall report offline conditions for cameras and sensors after the configured health-check threshold is exceeded.

8. System Modes and States

Mode / state	Description
Disarmed	Status and safety monitoring continue, but burglary alarm rules are inactive.
Armed Stay	Perimeter security is active while selected interior zones may be bypassed.
Armed Away	Perimeter and configured interior intrusion sensors are active.
Alert	A security or safety event is being evaluated or presented to the user.
Alarm	A qualifying alarm condition is confirmed; local alarm and notifications are active.
Maintenance	Authorized maintenance is allowed and affected devices may be temporarily excluded with clear indication.
Offline / Degraded	Internet/cloud connectivity is unavailable; essential local functions continue while remote functions may be limited.

SysML link
The companion SysML Draw.io file contains a dedicated state-machine diagram and an intrusion-event sequence diagram, in addition to the required block and use-case views.

8.1 Physical Requirements

Physical requirements

- **PHY-01 -** The central control unit shall be suitable for indoor residential installation near typical home-network equipment.
- **PHY-02 -** Outdoor cameras shall use enclosures rated for the intended exposure.
- **PHY-03 -** Battery-powered sensors shall allow battery replacement without destructive disassembly.
- **PHY-04 -** Each registered physical device shall have a unique label or scannable identifier for installation and maintenance.

8.2 Adaptability Requirements

Adaptability requirements

- **AD-01 -** Additional compatible sensors shall be addable without redesigning the central control architecture.
- **AD-02 -** The architecture shall support future expansion to at least 100 connected smart-home devices, subject to radio and network capacity.
- **AD-03 -** Device-adapter logic shall be isolated so that new supported protocols can be introduced with limited impact on core security rules.
- **AD-04 -** Storage capacity shall be expandable or configurable as video and event-retention needs change.

8.3 Environmental Conditions

Environmental requirements

- **ENV-01 -** Indoor electronic components shall operate in a nominal residential temperature range of 0 C to 40 C.

- **ENV-02 -** Outdoor cameras shall be selected for the temperature, moisture, dust, and precipitation exposure of their installation location.
- **ENV-03 -** Battery-powered sensors shall continue to operate during household AC-power interruption.
- **ENV-04 -** The control unit and critical network equipment should use backup power where required by the availability objective.
- **ENV-05 -** The system shall tolerate normal residential radio interference using supported retry, mesh-routing, or reconnection behavior.

8.4 System Security

Security requirements

- **SEC-01 -** Remote communication carrying credentials, commands, or private security data shall be encrypted in transit using current secure transport.
- **SEC-02 -** Passwords shall not be stored as plaintext.
- **SEC-03 -** Role-based authorization shall separate administrator and resident capabilities.
- **SEC-04 -** Administrative configuration shall be restricted to authorized administrator accounts.
- **SEC-05 -** Authentication failures shall be logged with timestamp and relevant non-secret context.
- **SEC-06 -** Security-significant actions such as arm/disarm, alarm activation, device removal, and administrator changes shall be auditable.
- **SEC-07 -** Supported devices shall use secure update processes and reject unauthorized update packages where capability permits.
- **SEC-08 -** Sensitive local data shall be protected by operating-system and application access controls; stored secrets and private cloud tokens should use encryption at rest.
- **SEC-09 -** Camera feeds and event data shall be limited to the permissions of the authenticated user role and household policy.

9. Verification

Verification checks whether the system design is built according to this specification. The companion verification/validation workbook contains executable test cases with objectives, priorities, preconditions, numbered steps, an Expected outcome immediately after every step, final acceptance criteria, and requirement IDs.

Requirement group	Primary verification method	Acceptance evidence
Motion / door / leak sensors	Functional integration test	Control unit receives the correct event and the expected log/alert behavior occurs.
Cameras	Demonstration + functional test	Authorized live view succeeds and camera availability is observable.
Authentication / authorization	Security functional test	Valid login succeeds, invalid login fails, and role restrictions are enforced.
Performance	Timed performance / load test	Measured event, alarm, and user-concurrency results meet PR thresholds.
Offline reliability	Fault-injection test	Internet is disconnected; local detection, alarm, and event retention continue.
Usability	Observed user test	Task-completion and interaction-count criteria meet the UR thresholds.
Security interfaces	Inspection + test	Remote interfaces use encrypted transport and access controls/logging are present.
Maintainability	Demonstration / timed maintenance test	Sensor replacement and diagnostics satisfy maintainability targets.

10. Assumptions and Dependencies

Assumptions and dependencies

- **A-01 -** Compatible sensors, controllers, cameras, and network equipment are installed and configured according to manufacturer instructions.
- **A-02 -** Remote access and push notifications depend on Internet connectivity and availability of the selected remote service.
- **A-03 -** Essential local intrusion detection and alarm operation do not depend on Internet connectivity.
- **A-04 -** Mobile and web users have supported devices, current software, and appropriate connectivity.
- **A-05 -** Household users maintain device batteries and replace them when low-battery status is reported.
- **A-06 -** LAN bandwidth and storage are sufficient for the configured camera count, resolution, and retention period.
- **A-07 -** Temporary cloud failure may remove remote functions but shall not disable local security behavior.
- **A-08 -** Installation locations provide adequate radio coverage, wired-network access, power, and environmental protection for the selected devices.

Appendix A - Requirement Traceability Summary

Requirement area	Representative IDs	Mapped test cases
Motion detection	FR-MOT-01 to FR-MOT-04	VER-001, VAL-001
Door/window	FR-DW-01 to FR-DW-03	VER-002, VAL-002
Water leak	FR-WL-01 to FR-WL-03	VER-003
Camera / remote view	FR-CAM-01, FR-CAM-02, FR-CAM-05	VER-004, VAL-003
Authentication	FR-AUTH-01 to FR-AUTH-05, SEC-03 to SEC-05	VER-005, VER-006
Local outage behavior	FR-CCU-03, REL-01, REL-02, REL-04	VER-007, VAL-008
Arm/disarm usability	UR-01, UR-02, UR-04	VAL-004, VAL-009
Load / concurrency	PR-01, PR-06	VAL-005, VAL-006
Alarm timing	PR-02	VAL-007

References

- ISO/IEC/IEEE 29148:2018, Systems and software engineering - Life cycle processes - Requirements engineering. The project uses the SysRS structure identified in section 9.4.
- NISTIR 8259A, IoT Device Cybersecurity Capability Core Baseline.
- NIST SP 800-213, IoT Device Cybersecurity Guidance for the Federal Government.

Project note

Connectivity choices in this educational design are architecture-level decisions. A production deployment would still require device-specific interoperability, regulatory, privacy, and security validation.